

Honeybees, caterpillars, butterflies, and other insects feed on a male flower's nectar, picking up pollen in the process, and then pollinating female flowers when they feed on them.

Red tide:

The harmful and toxic blooms of red Dinoflagellates causes the colour of sea water to become red. Red Dinoflagellates (Example: Gonyaulax) undergo such rapid multiplication that they make the sea appear red

Characters of Phycomycetes:

The mycelium is aseptate and coenocytic. Asexual reproduction takes place by zoospores (motile) or by aplanospores (nonmotile). These spores are endogenously produced in sporangium.

Examples of Phycomycetes:

- Mucor
- Rhizopus (the bread mould)
- Albugo (the parasitic fungi on mustard).

Eutrophication Too much nutrients causes harmful algal blooms (HABs) which are extremely toxic to living creatures (fish, shellfish, birds and mammals) It decreases the oxygen content rapidly

Aquatic biomes: Aquatic biomes are further classified into, Saltwater biome Freshwater biome

Terrestrial biomes: Tundra biome Desert biome Grassland biome Forest biome

Species whose populations have expanded dramatically and outcompete or displace native species, potentially threatening the structure and function of intact ecosystems. Non native species that have a potential to cause economic or environmental harm to humans are termed **as invasive species**

- The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake
- The recent illegal introduction of the African catfish *Clarias gariepinus* for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

Global warming:

The average temperature of earth surface is increasing day by day due to increase in concentration of greenhouse gases in the atmosphere. Human activities are main cause of it. The earth is gradually heating up due to rise in temperature. This phenomenon is called Global warming. It is causing serious problems to the health of plants and animals and also the surrounding environment.

Causes of Global warming:

Burning of fossil fuels Volcanic eruption Industrial effluents Deforestation Transportation Power plants

Effects of Global warming Rise in temperature Rise in Sea Level Climate change Extinction Droughts Loss of biodiversity

1. Protozoans:-

All protozoans are heterotrophs and live as predators or parasites. Amoeboid protozoans Flagellated protozoans Ciliated protozoans Sporozoans

Amoeboid protozoans: These organisms live in fresh water, sea water or moist soil. They move and capture their prey by putting out pseudopodia (false feet) as in Amoeba. Marine forms have silica shells on their surface. Some of them such as Entamoeba are parasites.

Flagellated protozoans: Freelifving or parasitic. They have flagella. The parasitic forms cause diseases such as sleeping sickness.

Ciliated protozoans: Aquatic and movement by cilia. A cavity (gullet) opens to outside of the cell surface. The coordinated movement of rows of cilia causes the water loaded with food to be driven into the gullet. Example: Paramecium.

Sporozoans: This includes diverse organisms that have an infectious sporelike stage in their life cycle. The most notorious is Plasmodium (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

Growth forms of Lichens :

Crustose :A type of lichen that form a flat crust on the substratum and firmly attached to substratum.

Foliose: This type of lichens has flat leaf like structures and are attached to substratum by thread like structures .

Fruticose: This type of lichens have erect filamentous like outgrowths and possess vertical pattern of growth.

Benefits of Lichens Bioindicators Ecosystem sustainability Medicines Animal food Dye for clothes

Common form of symbiosis are Commensalism , Mutualism.

Euglenoids: have a protein rich layer called **pellicle** which makes their body flexible.

Certain marine algae produce large amounts of volatile **DMS** (dimethyl sulfide). In the atmosphere, DMS is converted to sulfur dioxide, some of which in turn is converted to sulfur trioxide gas and to tiny droplets of sulfuric acid (H_2SO_4).

Neurospora is used extensively in biochemical and genetic work. Many members like morels and truffles are edible and are considered delicacies.

Importance of biodiversity ♣ ♣ ♣ Biodiversity performs important biological services.

1. Regulation of Climate
2. Soil formation and protection
3. Water resources protection
4. Education and scientific value
5. Recreation and ecotourism
6. Pollution breakdown and absorption
7. Economic value

2. Lentic water bodies

Standing bodies of freshwater that supports variety of life. Wind action is very vital in these water bodies because it increases the content of dissolved oxygen that is necessary for the aquatic life in these water bodies.

Effects of deforestation: Increase in carbon dioxide concentration in the atmosphere. Habitat destruction. Climate change. Loss of biodiversity. Hydrologic cycle disturbances. Soil erosion. Desertification.

3. Steps in nitrogen cycle

Fixation:

Two natural processes convert or fix N_2 into compounds useful as nutrients for plants and animals. Electrical discharges, or lightning, taking place in the atmosphere. Nitrogen fixation by nitrogen fixing bacteria

Nitrification:

Specialized soil bacteria convert most of the Ammonia (NH_3) and Ammonium ions (NH_4^+) in soil to nitrate ions (NO_3^-) which are easily taken up by the roots of plants.

Assimilation:

The plants then absorb nitrates from the soil into their roots to produce various amino acids, proteins, nucleic acids, and vitamins.

Ammonification:

When a plant or animal dies, specialized decomposer bacteria convert this detritus into simpler nitrogen-containing inorganic compounds such as Ammonia (NH_3) and water-soluble salts containing Ammonium ions (NH_4^+)

Denitrification:

Specialized bacteria convert (NH₃) and (NH₄⁺) back into nitrite and nitrate ions, and then into nitrogen gas (N₂) and nitrous oxide gas (N₂O). These gases are released to the atmosphere to begin the nitrogen cycle again.

4. The water cycle

is powered by energy from the sun and involves three major processes, Evaporation
Transpiration Precipitation.

Evaporation:

Evaporation changes liquid water into water vapor in the atmosphere. Incoming solar energy causes evaporation of water from the oceans, lakes, rivers, and soil.

Transpiration:

Over land, about 90% of the water that reaches the atmosphere evaporates from the surfaces of plants through a process called transpiration.

Precipitation:

Precipitation returns the water back to earth's surface. Most precipitation becomes surface runoff that flows into streams and lakes and eventually in to the oceans. From oceans it can evaporate to repeat the water cycle.

Strategy to Protect Ecosystems:

- Map global ecosystems and create an inventory of the species contained in each of them and the ecosystem services they provide
- Locate and protect the most endangered ecosystems and species, with emphasis on protecting plant biodiversity and ecosystem services
- Seek to restore as many degraded ecosystems as possible.
- Make development biodiversity friendly by providing significant financial incentives (such as tax breaks and write-offs) and technical help to private landowners who agree to help protect endangered ecosystems.

Solid waste management:

All waste that we generate can be categorized into three types, **Recyclable Biodegradable, Non-biodegradable.**

Domestic wastes are collected in small bins, which are then transferred to community bins by private or municipal workers. From these community bins, these are collected and carried to the disposable site. At the site, garbage is sorted out and separated into biodegradable and non-biodegradable materials. Non-biodegradable materials such as plastic, glass, metal scraps etc. are sent for recycling. Biodegradable wastes are deposited in landfills and are converted into compost.

Cryptic coloration (L. crypticus, hidden) :

Is a type of camouflage that occurs when an animal takes on color patterns in its environment to prevent the animal from being seen by other animals.

The color pattern of this tiger (*Panthera tigris*). Praying mantises sit in flowers of a similar color and ambush visiting insects.

Bacillus: These are rod shaped bacteria.

Biodiversity vary with respect to changes in the following factors, Climate change Geology Natural selection Species evolution Sea level.

Chemical warfare refers to use of chemical substance as a weapon for defence purpose.

Some predators use chemical warfare to attack their prey. For example, spiders and poisonous snakes use venom to paralyze their prey and to deter their predators.

5. Types of biodiversity over spatial scale

Robert Whittaker (1972) created a system to describe biodiversity over different spatial scales.

- Alpha diversity
Alpha diversity refers to the diversity within a particular area or ecosystem, and it is usually expressed by the number of species in that ecosystem. This is equivalent to measuring the species richness of an area. **Alpha diversity** is just the **diversity** of each site (local species pool).
- Beta diversity Beta **diversity** represents the differences in species composition among sites.
- Gamma diversity. Gamma **diversity** is the **diversity** of the entire landscape (regional species pool).

Seral stage

Each successional stage is called a seral stage.

Sere

The entire successional sequence is termed as a sere. Biomass increases, nutrients are conserved more efficiently, and productivity declines in the course of succession.

Communities begin to establish in areas which are devoid of life. The first community to become established in an area is called **the pioneer community**. The organisms that develop this pioneer community are called **pioneers**.

Climax community: The relatively stable and uniform community developed by the pioneers through different successional stages is named as climax community. It is a final and mature stage of any community.

Characters of Deuteromycetes:

- The Deuteromycetes reproduce only by asexual spores known as conidia.
- The mycelium is septate and branched
- Some members are saprophytes or parasites while a large number of them are decomposers of litter and help in mineral cycling.
- Examples of Deuteromycetes: Alternaria, Colletotrichum, Trichoderma.

6. On the base of production ozone is divided into two types:

Stratospheric ozone: In the first step, solar ultraviolet radiation breaks apart one oxygen molecule (O_3) to produce two oxygen atoms. In the second step, each of these highly reactive atoms combines with an oxygen molecule to produce an ozone molecule (O_3)

Tropospheric ozone: Nitrogen compounds released from automobiles is broken down by sunlight into nitrogen oxide (NO_2). It is then converted into nitrogen monoxide (NO), releasing an oxygen molecule that combines with oxygen (O_2) in the atmosphere to form Ozone (O_3).

Mimicry (L. mimus, to imitate) occurs when a species resembles one, or sometimes more than one, other species and gains protection by the resemblance. These six species of Heliconius are all distasteful to bird predators. A bird that consumes any member of the six species is likely to avoid all six species in the future.

Basidiomycete: The mycelium is branched and septate. The asexual spores are generally not found, but vegetative reproduction by fragmentation is common. The sex organs are absent, but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. The resultant structure is dikaryotic which ultimately gives rise to basidium.

Examples of Basidiomycetes: Agaricus (mushroom) Ustilago (smut) Puccinia (rust fungus)

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7. Carbon cycle:

- Carbon enters the atmosphere as **carbon dioxide** from respiration (breathing) and combustion (burning).
- Carbon dioxide is absorbed by producers (life forms that make their own food e.g. plants) to make **carbohydrates** in photosynthesis. These producers then put off oxygen.
- Animals feed on the plants. Thus passing the carbon compounds along the food chain. Most of the carbon these animals consume however is exhaled as **carbon dioxide**. This is through the process of respiration. The animals and plants then eventually die.

- The dead organisms (dead animals and plants) are eaten by **decomposers** in the ground. The carbon that was in their bodies is then returned to the atmosphere as carbon dioxide. In some circumstances the process of decomposition is **prevented**. The decomposed plants and animals may then be available as **fossil fuel** in the future for combustion.

Characteristics of protists, Aquatic habitat Unicellular eukaryotes Flagella or cilia Cyst formation Autotrophic or heterotrophic. Protists reproduce asexually by the process of cell fusion and sexually by a process involving zygote formation.

Classification of protists: Chrysophytes Dinoflagellates Euglenoids Slime molds Protozoans

Autogenic succession In autogenic succession the successional process is controlled by living or biotic component of an ecosystem **Allogenic succession** The entire successional process is controlled by non living or abiotic components of ecosystem.

Direct exploitation: Direct exploitation ranges from commercial activities such as logging operations or trade in endangered species to subsistence hunting.

Indirect exploitation: Indirect exploitation encompasses the unintentional mortality of non target species such as fish or turtles killed as bycatch in fishery operations. Both endanger species around the world

8. Components of biodiversity Species diversity Population diversity Community diversity Ecosystem diversity Genetic diversity Biogeographic diversity Functional diversity

Greenhouse effect:

Carbon dioxide released from burning fuels accumulates in the atmosphere and prevent the escape of heat coming from earth. This effect is termed as Greenhouse effect.

Causes of Greenhouse effect Human activities such as, Burning fossil fuels Deforestation Growing crops release carbon dioxide, methane, and nitrous oxide into the atmosphere.

ElNino effect:

Another type of shortterm climate change is the socalled El Nino effect, which occurs every 2 to 7 years around Christmas time, when the typical weather pattern in the Pacific Ocean breaks down. The trade winds weaken, and low pressure establishes itself in the central Pacific. It causes the winds to blow into the Pacific Basin from the west.

Types of nitrogen fixing bacteria

- Azobactor Free living aerobic bacteria
- Clostridium Free living anaerobic bacteria
- Chlorobium Photosynthetic bacteria
- Thiobacillus Chemosynthetic bacteria

- Nostoc Blue green algae

The process that involves the cyclic movement of nutrients between biotic (living portion) and abiotic (non living) environment of earth is termed as **biogeochemical cycle**. Matter is constantly recycled within ecosystems. Approximately 97% of living matter is made of oxygen, carbon, nitrogen, and hydrogen

Water pollution refers to any undesirable change in the quality of water that makes its consumption unfit for human and other animals. Polluted water has serious impact on health of living organisms.

Sources of water pollution Domestic waste Industrial waste Untreated sewage Oil spills Agricultural runoff Suspended or particulate matter.

Aposematic coloration

First, if it is small and strikingly beautiful, it is probably poisonous. Second, if it is strikingly beautiful and easy to catch, it is probably deadly

When wastes and poisons enter food webs, organisms at the highest trophic levels usually suffer the most. The accumulation of matter in food webs is called **biological magnification**